LOCAL BUSINESS SUPPORT RECOVERY

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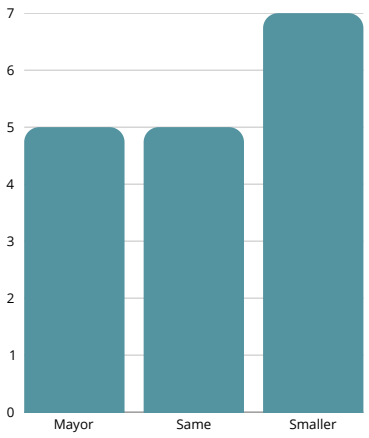
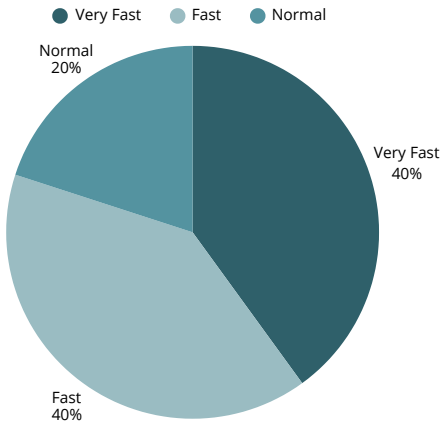
INTRODUCTION

After the pandemic, customer behavior at Titan Plaza shifted significantly. 41.18% of visitors now spend less than before, and foot traffic varies widely depending on the day. Small businesses struggle to keep up, facing inventory issues and low visibility compared to large chains. This platform helps them recover through data-driven engagement and inventory optimization.

RESULTS

Customer spending dropped significantly after the pandemic: 41.18% of visitors now spend less than before. This highlights the need for engagement strategies like personalized promotions and loyalty programs to encourage spending.





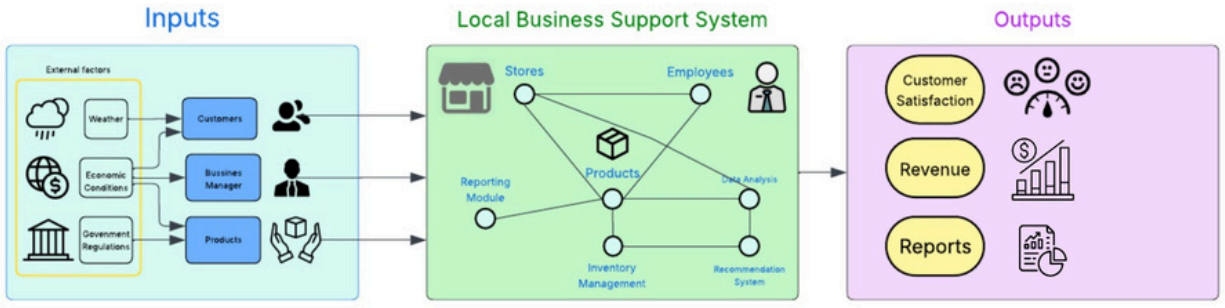
On the inventory side, 80% of small stores report fast or very fast rotation, meaning products move quickly. This makes inventory alerts essential to prevent stockouts during peak hours and ensure popular items are always available.

GOALS

Design a platform that helps small businesses increase customer loyalty and reduce stockouts by predicting foot traffic, sending automated promotions, and providing real-time inventory alerts.

PROPOSED SOLUTION

The platform integrates external factors like weather with customer and business data. The core system manages inventory and products, while a recommendation engine helps owners make data-driven decisions. Outputs include revenue reports and operational insights that improve engagement and inventory control.



EXPERIMENTS

We conducted a systems analysis at Titan Plaza using three methods:

- Direct observation: Foot traffic was counted on a Saturday (9am–12pm) and a Sunday (12pm–3pm) in the small stores area.
- Surveys: 17 customers and 5 employees were interviewed about spending habits, visit motives, and inventory management.
- Environmental correlation: Weather conditions and day of the week were recorded to understand their effect on foot traffic.

CONCLUSION

The platform directly addresses the challenges identified: personalized promotions and loyalty programs tackle the 41.18% drop in customer spending, while inventory alerts prevent stockouts in stores where 80% of products move quickly. By giving small businesses real-time data, the system helps them compete with larger chains and adapt to post-pandemic consumer behavior.

REFERENCES

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